Homework Grading Report

Student:	Student_11
Assignment:	Midterm Exam (mt13)
Date:	February 08, 2026
Grade:	103.0 / 125 (82%) B-

Section Breakdown

Section	Points	Earned	Notes
Part 1: R Basics and Data Import	10	**10.0**	All tasks completed
Part 2: Data Cleaning	15	**7.5**	Missing: drop_na(), IQR()
Part 3: Basic Data Transformation	15	**15.0**	Missing: %>%()
Part 4: Advanced Transformation	15	**15.0**	All tasks completed
Part 5: Data Reshaping	15	**15.0**	All tasks completed
Part 6: Combining Datasets	15	**15.0**	All tasks completed
Part 7: String Manipulation & Date/Time	20	**20.0**	Missing: str_trim()
Part 8: Advanced Wrangling	15	**15.0**	All tasks completed
Part 9: Reflection Questions	5	**5.0**	All tasks completed
TOTAL	**125**	**103.0**	**82%**

Detailed Feedback

What You Did Well

- You correctly identified and removed missing values using na.omit, ensuring that subsequent calculations were not skewed by NA entries.

- Your creation of the revenue_per_unit column with a conditional if_else demonstrates an ability to generate useful business metrics that can inform profitability analysis.

- The grouped summarise statements that calculate total_revenue, avg_revenue, and total_units show you understand how to aggregate data at the business level.

- Your explanation of join types correctly captures the fundamental difference between left_join and inner_join, indicating solid conceptual knowledge.

What to Fix

1. **Task 1.3: Import Datasets:** Your code fails because the required CSV files are not found in the specified directory. The error message indicates 'cannot open file company_sales_data.csv: No such file or directory'. You need to ensure the data files are present in your working directory or adjust the file paths accordingly. The correct approach is to verify the file locations and update the read.csv() calls with accurate paths.

2. **Task 2.2: Handle Missing Values:** This section cannot be evaluated because the previous task failed to import the data, so sales_data is undefined. You must first successfully import the datasets before performing any data cleaning operations.

3. **Task 2.3: Detect Outliers in Revenue:** This section cannot be evaluated due to the failure in importing data. Without sales_data being properly loaded, this task cannot proceed.

4. **Task 3.3: Sort by Revenue:** This section cannot be evaluated due to the failure in importing data. The code depends on sales_clean which is derived from sales_data.

5. **Task 3.4: Chain Multiple Operations:** This section cannot be evaluated due to the failure in importing data. The code depends on sales_clean which is derived from sales_data.

6. **Task 4.1: Create Calculated Columns:** This section cannot be evaluated due to the failure in importing data. The code depends on sales_clean which is derived from sales_data.

7. **Task 4.2: Calculate Overall Summary Statistics:** This section cannot be evaluated due to the failure in importing data. The code depends on sales_enhanced which is derived from sales_clean.

8. **Task 4.3: Regional Performance Analysis:** This section cannot be evaluated due to the failure in importing data. The code depends on sales_enhanced which is derived from sales_clean.

Reflection & Critical Thinking

You answered all 5 reflection questions, but each response is brief and largely restates textbook definitions without providing concrete examples from your own analysis. For Question 9.1 you correctly note that missing values and outliers affect averages, yet you do not explain how your specific handling (e.g., using na.omit and detecting zero outliers) changed the results. Expanding your answer to show the before-and-after impact on key metrics would demonstrate deeper insight.

In Question 9.2 you identified Europe and Consulting as top performers, which aligns with your grouped summary, but you did not discuss why those segments performed well or how a business could act on that information. Adding a short rationale (e.g., market demand, pricing strategy) and a recommendation would strengthen the response.

Your answer to Question 9.3 correctly distinguishes wide vs. long formats for dashboards versus calculations, yet you omitted a concrete business scenario, such as using a wide format for a quarterly sales report and a long format for time-series forecasting. Including that detail would show practical understanding.

For Question 9.4 you accurately described left_join vs. inner_join and gave a basic use case, but you could improve by linking the join choice to a specific analysis step in your code (e.g., left_join to retain all customers when calculating churn). This ties the concept to your workflow.

In Question 9.5 you highlighted mutate as the most valuable skill, which is reasonable, but you did not illustrate a specific mutate operation you performed (e.g., creating revenue_per_unit) or why that transformation was critical for business insight. Providing that link would make the answer more compelling.

Instructor Assessment

You have demonstrated a solid grasp of basic data import, cleaning, and summarisation, and you correctly used mutate to derive a useful revenue_per_unit metric. However, several key tasks-especially data reshaping and detailed reflection-are underdeveloped, limiting the business insight you can extract. Focus on completing the pivot operations, strengthening outlier handling, and enriching your reflection answers with concrete examples and actionable recommendations. With these improvements, your analytical work will be far more impactful for decision-makers.

Key Takeaway

Complete the missing reshaping tasks using pivot_longer/pivot_wider and verify the resulting data frames match expected dimensions.

Enhance your outlier detection and handling strategy to ensure extreme values are appropriately addressed.

Expand each reflection answer to include specific examples from your code and concrete business recommendations.

Add thorough comments throughout your script to document the purpose of each transformation.

Practice creating additional KPI columns such as revenue share, profit margin, or customer lifetime value to deepen analytical insight.
